Technische Universitaet Dresden – AnyWi Collaboration Opportunity

Summary from the call 07.04.2025

- As part of the AirMetro project, the Deutsche Telekom Chair of Communication Networks aims to develop a reliable multi-link communication infrastructure (cooperative communication) for UAVs operating in BVLOS. The research explores 5G networks and advanced algorithms, including network slicing and Time-Sensitive Applications Protocol, to meet the specific needs of UAV operations.
- AnyWi specializes in resilient communication and software infrastructure for mobile vehicles. Notable projects, such as COMP4Drones and ADACORSA - R&D efforts that facilitate BVLOS UAV operations was introduced and discussed.
- It has been identified that latency and 5G base station LOS interference are critical issues that need to be addressed.

Potential Topics for collaboration

- A **unified software platform** to handle heterogenous multi-link (5G, Satcom, Remote id etc..). This platform will enable the remote pilot to monitor link loss, and effective resource utilization during the flight operation
- **Network Slicing** for SAIL IV UAVs in 5G network. Here, we explore some **5G core network functions** enable traffic prioritization. A sidestep to see other potential similar to URLLC
- **Tactile Internet** principles for Human-in-the-loop for UTM infrastructure [1]
- **Ad-hoc networking**, **time-sensitive protocols** to create emergency UAV communication channel (in different technologies) – latency issues
- **Network security** (from UAS and UTM perspective) – Post-Shannon, Quantum key-based solutions [2]
- **Similar to V2X** - system architecture for UAVs
- **NTN/ 6G networks** - **Anticipative, Semantic communication** networks [3]
- Identify and enable solutions for **LOS Base station interference** (interference testing)

Projects and potential industrial and academic partners from Germany

General idea - https://cn.ifn.et.tu-dresden.de/research/projects/

Further direct point of Contacts

- Deutsche Telekom – T-Labs
- Meshmerize GmbH https://meshmerize.net/
- Atmosphere GmbH https://store.atmosphere.aero/index.php?id_category=16&controller=category
- Airbus Wireless Communication
- Ericsson
- Fraunhofer institute

Relevant Reference

1. Tactile Internet – CeTI – https://ceti.one/
   https://tu-dresden.de/tudresden/profil/exzellenz/exzellenzcluster/ceti?set_language=en
2. Research topics at Deutsche Telekom Chair - https://cn.ifn.et.tu-dresden.de/research/
3. Network testbeds - https://cn.ifn.et.tu-dresden.de/research/testbeds/